



Long-term impacts of school nutrition: Evidence from China's school meal reform

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ABSTRACT

Child malnutrition remains a major public health concern, especially in many developing countries. This paper examines the long-term effects of a school meal program on individuals' cognitive and health outcomes in rural China. Exploiting the staggered implementation of the Student Nutrition Improvement Program (SNIP), we find that early exposure to the SNIP (ages 6–15) significantly improved children's cognitive and health outcomes in the long run. These effects are not observed in urban residents who experienced similar socioeconomic changes in the same county. Empirical tests suggest that short-term improvements in children's nutrient intakes and mothers' labor market performance were likely to be potential mechanisms. Moreover, the effects of early exposure to the SNIP are stronger among children from low-socioeconomic status families. Our results suggest that the school nutrition program could be an important investment in children's long-run human capital. This may be relevant for many developing countries today that attempt to provide or have provided subsidized school meals.

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1. Introduction

Child malnutrition remains a major public health concern, especially in many developing countries. According to a report issued by the United Nations Children's fund in 2019, there are 200 million children under five suffering from stunting or wasting and 340 million children facing deficiency of essential vitamins and minerals.¹ The burden of child malnutrition primarily stems from poor-quality diets: about two thirds of children in the world do not get the minimum recommended dietary diversity for healthy growth and development (UNICEF, 2019). To improve children's nutrition, many countries have implemented school meal programs. Although a large body of studies has evaluated the short-term effects of school meal programs, few empirical studies have attempted to investigate their long-term effects (Bütikofer, Mølland, & Salvanes, 2018; Hinrichs, 2010; Lundborg, Rooth, & Alex-Petersen, 2021).

This study aims to examine the long-term impacts of China's Student Nutrition Improvement Program (SNIP) on children's health status and cognitive outcomes. Rural children usually have

worse health and nutritional status than urban children in China (H. Liu, Fang, & Zhao, 2013). To improve children's nutrition in rural areas, the Chinese government initiated the SNIP program in 2012. The program provided subsidized school meals to rural students in the compulsory education stage. It expanded rapidly and covered 40 million rural students in 1,762 counties in 2019. The regional and temporal variations in the implementation of the SNIP program allow us to adopt a difference-in-differences (DID) approach.

Using data from the 2018 wave of China's Family Panel Studies (CFPS), we show that early exposure to the nutrition program improved health status and cognitive ability in the long run. Specifically, being exposed to the reform before age 16 increased the probability of reporting excellent health status by 7.2 percentage points. Although there was no obvious effect on the positive emotion index, early exposure to the reform reduced the composite index for negative emotion by 0.112 standard deviations. With respect to cognitive ability, early exposure to the reform increased word and math test scores by 0.337 and 0.195 standard deviations, respectively.

Our results are robust to several sensitivity tests. First, we show similar trends in adult outcomes before the adoption of the SNIP. Our main results still hold if we control for differential cohort trends associated with predetermined county characteristics. These exposure effects are not observed in urban residents who

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¹ The two areas with the largest number of children who are stunted or wasted are Asia (88.7 million) and Africa (61.0 million).

are not covered by the program. Then, we demonstrate minimal bias induced by selective migration of respondents. Our baseline estimates remain similar when we use alternative sample restrictions about respondents' migration experiences. Lastly, we verify that other policies that might have been running at the same time should not bias our estimates.

We claim that the relevant mechanism driving our results is enhanced nutritional status and improved family economic conditions during childhood. Using data from the 2010–2018 waves of the CFPS, we provide some evidence that the SNIP program immediately increased children's consumption of high protein foods (beans, grains, eggs, and milk), improved mothers' labor market performance, and raised household expenditure on education. We find no evidence that the school meal program affected body size, implying that the program did not promote unhealthy excessive calorie intake. Furthermore, we show that the effects of early exposure to the SNIP are more pronounced among children from low-socioeconomic status families, which is consistent with the mechanism of improved childhood nutrition.

This study contributes to the literature by establishing a link between improved nutrition in childhood and long-run health and cognitive outcomes. A growing body of literature has documented that government policies that improve conditions in utero and early childhood can have significant long-term effects for those exposed in utero and early childhood (Almond, Currie, & Duque, 2018; Currie & Almond, 2011).² So far, however, limited attention has been paid to the effects of health and nutrition intervention during later phases of childhood and adolescence. Our estimates point to large long-term beneficial effects of government-sponsored nutrition programs on health status and cognitive ability for school-age children. In addition, our results suggest an important role of school-based health intervention in reducing inequalities in health outcomes and cognitive skills. Children from low-income families usually bear a heavy burden of malnutrition and exhibit poor health status and sluggish cognitive development (Sorhaingo & Feinstein, 2006). Poor children gained larger benefits from the school meal program than wealthy children did. Consequently, the program could narrow socioeconomic disparities in adulthood.

The remaining part of the paper proceeds as follows. Section 2 discusses the related literature and institutional background. Section 3 describes the data and provides the descriptive statistics. Section 4 presents the estimation strategy. Section 5 shows the estimation results and analyzes the underlying mechanism. Section 6 offers a brief conclusion.

2. Related literature and background

2.1. Related literature

School meal programs exist in almost every country in the world (Drake et al., 2017). The evaluation of school meal programs has attracted considerable attention among scholars in the fields of medicine, public health, psychology, and economics. Most of the existing literature has revealed that the school meal programs increased the intake of calories and various nutrients (Alderman & Bundy, 2012; Jomaa, McDonnell, & Probart, 2011). Gleason and Suitor (2003) revealed that participation in the National School Lunch Program in the United States increased children's intake of vitamins, minerals, and dietary fiber. Bhattacharya, Currie, and

Haider (2006) found that the School Breakfast Program in the United States improved the nutritional quality of the diet. Afridi (2010) showed that a nationally mandated school meal program reduced hunger at school and protein-energy malnutrition in India.

A large body of literature has shown the beneficial effects of school meals on health outcomes in the short run (Jomaa et al., 2011). For example, Neumann, Murphy, Gewa, Grillenberger, and Bwibo (2007) found that meat and milk supplementation at school improved linear growth and weight gains in Kenya. Arsenault et al. (2009) showed that the provision of a school snack was negatively associated with morbidity among children in Colombia. Gundersen, Kreider, and Pepper (2012) found that the National School Lunch Program reduced the probability of reporting poor health among children in the United States. X. Liu, Zhao, and Chen (2019) observed that subsidized school meals significantly improved students' mental health status measured by their anxiety scale in rural northwestern China. Nevertheless, several studies have shown that school meals led to increased obesity rates among students in the United States (Millimet, Tchernis, & Husain, 2010; Schanzenbach, 2009).

A number of studies have documented positive effects of school meals on test scores in the short run (Anderson, Gallagher, & Ritchie, 2018; Figlio & Winicki, 2005; Frisvold, 2015; Imberman & Kugler, 2014). For example, Belot and James (2011) evaluated a program that improved nutritional standards of school lunches in the U.K. and found that the program increased English and science test scores among 11-year olds. Anderson et al. (2018) found that contracting with a healthy school-meal vendor increased test scores among students at public schools in California. Chakraborty and Jayaraman (2019) found that India's midday meal program increased math and reading test scores of primary school-aged children. However, several studies have shown limited beneficial effects of school meals on test scores (Imberman & Kugler, 2014; Leos-Urbel, Schwartz, Weinstein, & Corcoran, 2013; McEwan, 2013). For instance, McEwan (2013) assessed a school lunch program in Chile and found no obvious effects of higher-calorie meals on fourth-grade math and language test scores.

School meal programs can affect children's long-run outcomes through several potential channels (Sorhaingo & Feinstein, 2006). First, they can improve childhood health and facilitate physical growth. Nutrition affects physical growth more in the postnatal period than in the prenatal period. The most important nutrient for physical growth is protein, followed by minerals and vitamins A and D (Sorhaingo & Feinstein, 2006). School meal programs may increase children's nutrient intake, enhance their physical fitness, and reduce their risk of disease. Second, adequate nutritional intake can facilitate the development of cognitive skills (Drake et al., 2017). School meals contain essential nutrients, such as zinc, iodine, iron, and folate. These nutrients can directly improve short-term memory and learning abilities. School meals may also make students more attentive in class and raise their energy level. Third, school meals can affect outcomes through non-nutritional mechanisms. The provision of meals can affect the incentives to attend school and thereby increase attendance rates among students (Jayaraman & Simroth, 2015). School lunches might decrease household expenditure on food and ease the burden of household work, and thereby enable more parents to enter the labor market and increase family income. The increased household income might benefit children in the long run (Dahl & Lochner, 2012).

Despite its theoretical plausibility, there exists limited evidence regarding the long-term effects of school meal programs. Hinrichs (2010) assessed the long-run effects of the National School Lunch Program in the United States. He showed that subsidized school lunches generated no obvious long-term health benefits but had large and positive effects on educational attainment. Bütikofer et al. (2018) evaluated a free school breakfast program in Norwe-

² For example, previous studies have assessed the long-term effects of government intervention such as disease eradication programs (Baird, Hicks, Kremer, & Miguel, 2016; Bleakley, 2007); availability of alcohol and tobacco (Bharadwaj, Johnsen, & Løken, 2014; Nilsson, 2017); pollution reduction measures (Grönqvist, Nilsson, & Robling, 2020; Isen, Rossin-Slater, & Walker, 2017); and infant care (Bharadwaj, Løken, & Neilson, 2013).

gian cities. They observed that access to nutritious school breakfasts boosted educational attainment and adult earnings. [Lundborg et al. \(2021\)](#) examined the long-term effects of a school lunch program, which provided free nutritious lunches to all students in Swedish primary schools. They found positive effects of access to free lunches on height, early adult physical health, educational attainment, and lifetime income. Our paper aims to extend the literature on school meals by studying their long-term impacts on physical health, mental health, and cognitive performance.

2.2. Background

Although China has undergone remarkable economic and urban growth since its market-oriented reform in 1978, its rural areas remain plagued by poverty, weak infrastructure, and limited social support outside the immediate family unit. According to the CFPS community survey, in 2010 approximately 92.2% of the sampled villages had access to electricity but only 71.1% had access to satellite or cable TV services; 56.8% had medical facilities (hospitals/clinics/drug stores); 52.8% had access to tap water; 17.7% had enterprises run by local residents; and 1.5% had access to pipeline gas.

The school dining environment was unsatisfactory in rural areas. The school canteens usually had poor infrastructure and many of them lacked cold storage and disinfection facilities. According to a survey on canteens of rural boarding schools, 40% of the canteens did not provide breakfast, 50% of the canteens only provided staple foods for lunch and dinner, 50% of the canteens provided less than three kinds of fresh vegetables each day, and 27% of the canteens did not provide meat, 60% of the canteens did not provide eggs, and 85% of the canteens did not provide milk ([Zhou et al., 2015](#)). In addition, China's rapid development and urbanization induced many rural adults to migrate from their homes to urban areas in search of better job opportunities. According to a report released in May 2013 by China's Women's Federation, left-behind children reached 61 million, which accounted for 37.7% of rural children. The lack of caregivers may worsen the nutrition status of rural children ([Zhou et al., 2015](#)). Many children in rural areas could not eat nutritious meals when attending school. According to a survey conducted by the China Development Research Foundation Survey, about 47.6% of rural students could only eat two meals a day, and 66.4% of them could not eat breakfast.

Consequently, malnutrition was a major health concern among rural children. According to the 5th National Nutrition and Health Survey in 2012, the prevalence of stunting and underweight among children under six in poor rural areas was 19.1% and 5.1%, respectively. Micronutrient deficiency among rural children was also a common problem. The prevalence of anemia among children under six in poor rural areas was 16.6%. The prevalence of vitamin A deficiency among rural children aged between 6 and 12 years old was 32.9%.

In order to fight childhood malnutrition, some local authorities implemented some small-scale school meal programs such as the Egg and Milk Project, the Love Nutrition Meal Program, and the Free Breakfast Program. In 2011, the Chinese government initiated a nationwide school meal program for rural students in the compulsory education stage, namely the Student Nutrition Improvement Program (SNIP). According to the Progress in Nutrition Improvement of Students in Rural Areas in Poverty-stricken Areas published by the China Development Research Foundation, the central government has allocated 159.1 billion CNY to the NIP since its launch, covering 134,000 schools and benefiting 36 million students.

The school meal program feature contains several features ([Zhang, Hu, Tian, Zhang, & Ma, 2015](#)). First, the central government

provides meal subsidies to improve their nutrient intakes. The daily subsidy was typically three yuan per student before 2012 and rose to four yuan in 2014. To understand the extent to which the school meal program changed the nutritional quality of the school meals, we can compare their nutritional content before and after the reform. A typical lunch of poor rural students before the reform might include 150 g of rice, 100 g of potatoes, and 5 g of cooking oil. A three-yuan subsidy could substantially improve the lunch. The lunch after the reform might include 150 g of rice, 250 g of spinach, 50 g of meat, and 5 g of edible oil, and could be modified to include other types of vegetables, milk, and eggs. The lunch included nutrients such as high-quality proteins, essential minerals, and vitamins. Second, the central government provided financial support for schools to build and improve canteen facilities. The school canteens took a major role in serving meals and their improvements were an important step to ensure the provision of healthy and safe food. Third, local authorities established new departments to supervise food purchase and production. Schools were also required to disclose food-related information in public announcement systems.

3. Data

3.1. Reform timing

Information on the implementation of the school meal program is collected from official implementation plans, local yearbooks, and health gazetteers. Although the timing of the reform is guided by the central government, local governors could decide the implementation of the reform based on local socioeconomic conditions. The central government gave priority to counties in poor counties. Nevertheless, some local governments in rich areas, such as Shanghai and Zhejiang Province, adopted the reform early. [Figure A1](#) plots the frequency of the adoption years of the SNIP program for the 162 counties covered by the CFPS. A total of 79 counties have not adopted the SNIP program by 2020.

We formally compare the differences in baseline characteristics between treated and untreated counties in the CFPS sample. The county-level variables are extracted from the 2011 China County Statistical Yearbook and the 2010 county-level census data. [Table 1](#) reports the corresponding statistics. We find small and statistically insignificant differences between SNIP and non-SNIP counties in terms of the total population, sex ratio, proportion of people aged above 15, population growth rate, and years of education. Nevertheless, treated and untreated counties differed significantly in terms of the number of teachers, proportion of people aged below 15, employment rate, share of the urban population, and GDP per capita. Most of these differences suggest that SNIP counties were less economically developed than non-SNIP counties at baseline.

These systematic differences in program implementation could give rise to spurious estimates of the program effects if those same county characteristics are associated with differential trends in children's long-term cognitive and health outcomes. In order to control for possible differences in trends across counties that are spuriously associated with the SNIP effects, we include interaction terms of these pretreatment county characteristics with birth cohort dummies in the robustness check.

3.2. CFPS

The individual data comes from the CFPS, which is an ongoing household panel survey by Peking University in China. The survey has been conducted every two years since 2010. Respondents who left their initial family were still surveyed in the following wave. The follow-up rate of the survey reached approximately 80%. The

Table 1
Baseline differences between treated and untreated counties.

Variables	Untreated counties		Treated counties		Difference	p-value
	Mean	SD	Mean	SD		
	(1)	(2)	(3)	(4)	(5)	(6)
The log of total population	13.19	0.711	13.06	0.831	0.139	0.257
Sex ratio	105.1	6.794	104.7	5.778	0.441	0.656
Share of people aged above 65	8.946	2.160	9.459	2.417	−0.512	0.160
Share of people aged below 15	14.58	5.192	16.50	6.029	−1.914	0.034**
Population growth rate	3.356	3.498	4.073	3.852	−0.717	0.221
Share of the urban population	41.92	29.28	29.00	25.88	12.92	0.003***
Years of education	9.723	1.204	9.403	1.341	0.320	0.115
The log of GDP per capita	10.45	0.909	9.919	1.050	0.528	0.001***
The log of the number of teachers	6.516	0.881	6.233	0.955	0.283	0.053*
Employment rate	76.48	5.108	73.93	5.361	2.553	0.002***

Notes: The data is from the 2011 China County Statistical Yearbook and 2010 county-level census data. Columns (1) and (2) show sample mean and standard deviation of observable characteristics of treated counties in 2010, respectively. Columns (3) and (4) show sample mean and standard deviation of observable characteristics of untreated counties in 2010, respectively. Columns (5) and (6) show the differences between treated and untreated counties and their p-values. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

baseline survey was conducted in 2010 in 162 counties in 25 provinces, with a sample size of 8,438 households. The survey collected information on demographic characteristics, health outcomes, cognitive ability, nutrient intakes, migration experiences, and economic activities. Xie et al. (2017) documented that a series of demographic and socioeconomic variables in the CFPS dataset (e.g., age, ethnicity, gender, educational achievement, income, and health) shared similar characteristics with those in other nationally representative datasets.

Reform exposure — We construct three measures of reform exposure based on China's institutional settings. China's compulsory education generally includes six years of primary school and three years of middle school. Children usually started primary education at six. When the SNIP was implemented in the county, children under 16 were affected by the school meal program. Therefore, the first measure of reform exposure is a binary variable indicating whether the respondent was <16 years old when the SNIP was implemented in the county. This measure is simple, easy to interpret, and insensitive to uncertainty about the critical exposure period. It is sensible to assume that children exposed to the reform below the age of 16 are more likely to be affected by the school meal program in early life than those exposed to the reform above the age of 16.

We also quantify the duration of reform exposure using the number of years that the respondent was exposed to the school meal program at ages 6–15 (Fang & Chen, 2021; Xiao, Li, & Zhao, 2017). Individuals aged above 16 at the implementation of the SNIP in each county are never exposed to the school meal reform, with a corresponding exposure value of 0. Those aged 6–15 are partly exposed to the reform, with reform exposure defined as 16 minus their age at the implementation year. Those under 6 are considered as exposed during their entire period of compulsory education, with an exposure value of 9. Individuals with long reform exposure might receive large nutrition investments, thereby having good academic performance in class, and enjoying a relatively long period of good health status.

In addition, we set two dummy variables to represent different lengths of exposure. If a child is enrolled in primary school at age 6, he or she will complete primary school and start middle school at age 12. We classify individuals into three groups according to their age at first exposure to school meals: those aged under 12 are fully exposed cohorts, those aged 12–15 are partially exposed cohorts, and those aged 16 and older are non-exposed cohorts. We then set two dummy variables for being partially and fully exposed cohorts.

Cognitive outcomes — We use test scores to measure an individual's cognitive ability. The CFPS applied a series of verbal and

mathematical problems to assess cognitive ability of respondents. In the verbal test, the respondents were required to read a set of Chinese characters aloud. The final score was determined by the number and accuracy of the characters that the respondent could read. In the math test, the respondents were required to solve a set of math questions, including addition, subtraction, multiplication, division, logarithms, exponents, trigonometric functions, permutation, sequence, and combination. The math score was calculated from the number of questions that the respondents answered correctly.³

Health status — The 2018 wave of the CFPS asked all respondents to rate their health as “excellent, very good, good, fair, or poor”. Self-rated health status has been shown to be a robust and powerful predictor of morbidity and mortality (Wu et al., 2013). We recode the health status index so that one indicates poor status and five for excellent status. We then construct a dummy variable representing whether the respondents reported excellent health status.

The 2018 wave of the CFPS also contained several questions about mental health status. The respondents were asked to rate the frequency of certain feelings in the last month on a four-point scale. Specifically, the respondents evaluated (1) the frequency of feeling lonely, (2) the frequency of feeling sad, (3) the frequency of poor sleep, (4) the frequency of enjoying life, and (5) the frequency of feeling happy. We construct two indices: negative emotion and positive emotion. The composite index is the simple average across standardized z-score measures of each variable. The z-score is the number of standard deviations which a value of a variable is away from the mean. In the case of negative emotion, all components are negative (loneliness, sadness, and poor sleep) and an increase in the negative emotion index indicates a worse outcome. For positive emotion, both components are positive (enjoying life and happiness) and a higher score represents a better outcome. As shown by Kling, Liebman, and Katz (2007), aggregating multiple measures in a given field increases statistical power.⁴

Nutrition — The 2010–2014 waves of the CFPS collected information on individuals' nutrient intakes. The CFPS asked whether the respondents consumed certain categories of food in the last

³ One might doubt whether cognitive outcomes are comparable among individuals aged 17–29. Table A1 in the appendix reports the estimates for cognitive test scores standardized by age. Our main results remain qualitatively unchanged.

⁴ The 2018 wave of the CFPS provided a simplified version of the Center for Epidemiologic Studies Depression Scale (CES-D) to measure symptoms of depression. A high CES-D value stands for a high risk for clinical depression. We provide the estimate for the CES-D scale in Table A1 in the appendix. The estimated coefficient is negative but statistically insignificant.

week. We set four dummy variables for consuming fish and meat, eggs and milk, beans and grains, and vegetables and fruits, respectively. The 2010–2018 waves of CFPS also collected measures of self-reported height and weight. We calculate each respondent's body mass index (BMI) and then construct two dummy variables for being overweight (BMI greater than 23) and underweight (BMI < 18.5).

Family economic outcomes — The 2010–2018 waves of CFPS collected information on household expenditures on food and education. In addition, the survey gathered information on parents' employment status and annual income. We use these variables to assess the short-term effects of the SNIP reform on household economic outcomes.

3.3. Regression sample

In the empirical analysis, we use the 2018 wave of the CFPS to evaluate the long-term effects of the SNIP. We limit the sample to respondents aged between 17 and 29. Individuals aged 30 and above could be too old to be comparable with the treated cohorts. In rural areas, age misreporting, delayed school starting age, and grade retention were not uncommon. Consequently, respondents who were aged 16 when the SNIP was implemented might be affected by the SNIP. To avoid confusion, we exclude respondents who were aged 16 when the reform was implemented. We use children's residence county at the age of 12 to match with the county-level reform timing. Table 2 presents the summary statistics of baseline characteristics. The average respondent is about 23.4 years old and 52.5% are male. In terms of health characteristics, 24% reported excellent health status. In our regressions, the number of observations varies since the fraction of missing data varies somewhat across outcomes.

In the mechanism analysis, we use the 2010–2018 waves of the CFPS. When analyzing nutrition-related variables, we limit the sample to children aged 6–15 in rural areas. When analyzing family economic variables, we restrict the sample to families or parents with children aged 6–15 in rural areas. Table 3 presents the descriptive statistics for the main variables in the mechanism analysis. The average child is about 10.4 years old. Approximately 53.0% of the children are male, and 41.2% are exposed to the SNIP. About 77.8% of mothers with children aged 6–15 are working and 87.6% of fathers with children aged 6–15 are working. The sample sizes are different for each outcome variable due to data availability.

4. Identification strategies

We use a DID strategy to evaluate the effects of early exposure to the SNIP on cognitive and health outcomes. We compare cognitive ability and health outcomes for those with early childhood exposure to the SNIP in their county to those born earlier (and therefore without childhood exposure to the school meal program). We make the following assumption: in the absence of the school meal program, the changes in average outcomes between different cohorts in counties that implemented reform would have been the same as the changes for those residing in counties whose reform status kept unaltered between different cohorts. The school meal program induces a deviation from this parallel trend. In traditional DID models, panel data are used routinely and treatment status varies across space and time. By contrast, our model relies on cross-sectional data and exploits variations in treatment status across space and cohort. Although this strategy is slightly different from the traditional DID model, it has been widely adopted in previous studies (Yi Chen, Fan, Gu, & Zhou, 2020; Duflo, 2001).

Specifically, we estimate the following model:

$$Y_{icb} = \beta_1 \text{Exposure}_{cb} + X_{icb}\gamma + \alpha_c + \lambda_b + \varepsilon_{icb} \quad (1)$$

where i , c , and b denote individual, county of residence at age 12, and cohort, respectively. Y represents the outcomes of interest, including self-rated health status, cognitive ability, and mental health status. *Exposure* is a measure of early exposure to the SNIP. In our main analysis, it is a dummy variable indicating that an individual was not more than 15 years old when the SNIP was adopted in the county. In addition, we use an alternative measure which is the number of years exposed to the reform during ages 6 to 15. The parameter β_1 provides the effect on outcomes of an additional unit of early exposure to the SNIP.

We include county fixed effects (α_c) that account for persistent differences across counties over time and birth cohort fixed effects (λ_b) that control for nationwide cohort trends in the outcomes. To increase the statistical precision and test the sensitivity of our specification, the regressions also control for individual and parental background characteristics (X). Specifically, X includes gender, an indicator for Han ethnicity, parents' political status (indicators for community membership), and parents' years of education. ε_{icb} is the error term representing unobserved determinants of children's long-run outcomes. The standard errors are clustered at the county level to account for both spatially and serially correlated errors within the counties (Bertrand, Duflo, & Mullainathan, 2004).

In the mechanism analysis, we employ an alternative DID model to evaluate the short-term impacts of the SNIP. The model is specified as follows:

$$Y_{ict} = \beta_1 \text{Treatment}_{ct} + X_{ict}\gamma + \alpha_c + \lambda_t + \varepsilon_{ict} \quad (2)$$

where i indexes the individual, c the county of residence, and t the survey year. Y represents a range of short-term mechanism outcome variables, including nutrient intakes, parental labor supply, and household expenditures. In the analysis of children's nutrition-related outcomes, *Treatment* is a dummy variable indicating whether the child was exposed to the SNIP. In the analysis of family economic outcomes, *Treatment* is defined as the number of children affected by the SNIP. α_c represents county fixed effects and λ_t standards for survey year fixed effects. Again, ε_{ict} is the error term, and the standard errors are clustered at the county level.

5. Empirical results

5.1. Baseline estimates

Table 4 presents the estimates from our main specification (1). We measure early exposure to the SNIP by an indicator for being exposed to the reform below age 16. We begin by estimating the long-term impact of the school lunch program on self-reported health status. Column (1) shows that the introduction of the SNIP significantly improved self-rated health status for younger cohorts. In the regression sample, the average length of reform exposure conditional on receiving treatment is 3.2 years. The estimated coefficient in Column (1) indicates that for an average child, being exposed to the SNIP in childhood increased the probability of reporting excellent health status by 7.2 percentage points. This corresponds to a 30.1% increase at the mean. The size of the effect is large compared to the one estimated for school lunches in Sweden (Lundborg et al., 2021). They found that nine years of school lunch exposure raised the probability of reporting near perfect health by 4 percentage points, corresponding to a 6% increase at the mean.

We then turn to mental health outcomes. Columns (2) and (3) of Table 4 present the estimates for mental health outcomes. Although early exposure to the SNIP had no obvious effects on the composite

Table 2
Summary statistics of adult data (the 2018 CFPS).

	(1)	(2)	(3)	(4)	(5)
VARIABLES	N	Mean	Standard Deviation	Minimum	Maximum
Explanatory variables					
Male	4,151	0.525	0.499	0	1
Birth year	4,151	1,995	3.871	1,989	2,001
Age	4,151	23.42	3.871	17	29
Han ethnicity	4,151	0.858	0.349	0	1
Fathers' years of education	4,151	6.040	4.166	0	22
Mothers' years of education	4,151	4.495	4.090	0	22
Fathers' political status	4,151	0.0246	0.155	0	1
Mothers' political status	4,151	0.00313	0.0559	0	1
Dummy reform exposure	4,151	0.223	0.416	0	1
Linear reform exposure	4,151	0.710	1.522	0	6
Outcome variables					
Health status	4,105	3.639	1.006	1	5
Excellent health status	4,105	0.239	0.426	0	1
Negative emotion	3,519	0.0307	0.693	−0.703	3.262
Positive emotion	3,519	0.0238	0.779	−2.065	1.105
Word test score	2,027	26.20	7.401	0	34
Math test score	2,027	15.59	5.664	0	24

Table 3
Summary statistics of mechanism data (the 2010–2018 CFPS).

	(1)	(2)	(3)	(4)	(5)
VARIABLES	N	Mean	Standard Deviation	Minimum	Maximum
Child variables					
Survey year	15,011	2,014	2.891	2,010	2,019
Male	14,997	0.530	0.499	0	1
Birth year	15,008	2,003	4.202	1,995	2,014
Age	15,011	10.37	2.917	6	15
Han	15,011	0.836	0.370	0	1
Fathers' years of education	15,011	5.527	4.257	0	16
Mothers' years of education	15,011	4.152	4.126	0	16
Fathers' political status	15,011	0.0486	0.215	0	1
Mothers' political status	15,011	0.00606	0.0776	0	1
Treatment	15,011	0.412	0.492	0	1
Vegetables and fruits	5,123	0.931	0.254	0	1
Fish and meat	5,123	0.678	0.467	0	1
Milk and eggs	5,123	0.705	0.456	0	1
Beans	5,123	0.471	0.499	0	1
Underweight	13,498	0.633	0.482	0	1
Overweight	13,498	0.0919	0.289	0	1
Family variables					
Mothers' employment	8,644	0.778	0.416	0	1
Fathers' employment	8,321	0.876	0.329	0	1
Mothers' income	3,066	8.718	1.519	1.609	13.59
Fathers' income	4,368	9.629	1.170	1.386	13.42
The log of educational expenditure	10,967	6.981	2.435	0	11.43
The log of food expenditure	10,819	9.031	1.142	0	12.57
Number of treated children	11,142	0.5576198	0.7892844	0	4

Table 4
Baseline estimates.

	(1)	(2)	(3)	(4)	(5)
VARIABLES	Excellenthealth status	Negativeemotion	Positiveemotion	Wordtest score	Mathtest score
Reform exposure	0.072** (0.030)	−0.112** (0.052)	−0.019 (0.065)	2.497*** (0.865)	1.105** (0.494)
Observations	4,078	3,493	3,493	2,014	2,014
R-squared	0.067	0.069	0.080	0.330	0.370
Birth year FE	YES	YES	YES	YES	YES
County FE	YES	YES	YES	YES	YES

Note: The core independent variable is a binary variable indicating whether the respondent was exposed to the SNIP program before age 16. Other control variables include gender, Han ethnicity, parents' community membership, and parents' years of education. The standard errors are clustered at the county level. ***significant at 1% level, **at 5%, *at 10%.

index for positive emotion, it significantly decreased the composite index for negative emotion by 0.112 standard deviations.⁵ No previ-

⁵ We also report the separate estimates for mental health outcomes in Table A2. The estimates indicate that reform exposure significantly reduced the feelings of sadness and loneliness.

ous studies have evaluated the long-term effects of school meals on mental health. We can relate our findings to findings from other short-term evaluations of school meal programs, such as X. Liu et al. (2019) who found that a subsidized school meal program in Chinese schools reduced a composite anxiety index by 0.081 standard deviations.

Cognitive ability has been generally considered an important predictor of job performance. Columns (4) and (5) of Table 4 show the long-term effects of the SNIP reform on cognitive test scores. The estimates indicate that early exposure to the SNIP increased word by 2.497 points or 0.337 standard deviation and math test scores 1.105 points or 0.195 standard deviations. No previous study has examined the long-term effects of school meals on cognitive skills. The effects observed in this study are not small compared with the literature studying the short-term effects of school meal programs on test scores. For example, Imberman and Kugler (2014) showed that the implementation of a free in-class breakfast program in the Southwest United States increased math and reading test scores by 0.09 and 0.06 standard deviations, respectively. Frisvold (2015) revealed that the availability of the School Breakfast Program in the United States raised math and reading test scores by 0.292 and 0.362 standard deviations, respectively.

5.2. Robustness check

In this subsection, we perform a range of robustness tests, including checking for parallel trends prior to treatment assignment, a placebo test using the urban population, using an alternative measure of reform exposure, excluding the influence of other policies that might have been running at the same time, and examining the issue of selective migration. The main results remain robust and consistent.

Parallel trends – The validity of the DID estimates relies on the parallel trends assumption. Specifically, the differences between treatment groups and control groups would be similar over time in the absence of the SNIP reform. A common concern is that the DID estimates might confound the dynamic effects of the SNIP program with pre-existing differences in time trends across treated and untreated counties. In other words, individuals might have experienced improvement in cognitive ability and health outcomes due to differences in time trends that succeeded the SNIP reform. We adopt four approaches to tackle this issue.

First, we estimate a fully flexible event-analysis model. The equation takes the following form:

$$Y_{icb} = \sum_{k=8}^{22} Exposure_age_k \delta_k + X_{icb} \gamma + \alpha_c + \lambda_b + \varepsilon_{icb}, \quad (3)$$

where $Exposure_age_k$ is a dummy variable indicating whether the respondent was k years old when the SNIP reform was adopted in the county. Other variables are defined as in Equation (1). The omitted baseline group is the respondents who were more than 23 years old at the implementation of the SNIP. Therefore, the estimated coefficient δ_k tells us the difference in the cognitive skills and health outcomes of the group exposed to the SNIP reform at age k relative to the baseline group. This specification allows us to examine the relationship between age at initial rollout and future outcomes in a nonparametric way. Note that longer exposure always means earlier exposure to the school meal program in our setting. Therefore, we cannot distinguish the effect of the length of the exposure from that of age at first exposure to the school meal program. The school meal program shares this characteristic with other social safety net programs, such as the Moving to Opportunity Program and the Food Stamp Program (Chetty, Hendren, & Katz, 2016; Hoynes, Schanzenbach, & Almond, 2016).⁶

Figure 1 plots the estimated coefficients δ_k from Equation (3) as well as the 90% confidence interval. The variable on the x-axis is

the age of the respondent when the reform was adopted in the county. The graphs shown here are the inverse of a typical event study graph, in that treatment (exposure to the SNIP) increases as we move from the right side (treated in later life) to the left side (treated in early life). We learn two important facts from the fully flexible estimates. First, we do not observe any clear trends of the estimated exposure effects for the cohorts aged above 16. The estimated differences between the older exposed group and the baseline group are small and statistically not significantly different from zero. These cohorts had usually left middle schools at the implementation of the reform and could not benefit from the SNIP. The second insight we gain is that the cognitive skills and health outcomes of cohorts exposed to the reform before age 16 began to improve relative to the baseline cohort. The effects on word test scores appear to begin immediately at the threshold (i.e. age 15), while the effects on math test scores and health outcomes were less noticeable.⁷ Overall, the timing of the improvements in cognitive and health outcomes is consistent with our identification strategy.

Second, we allow the cohort effects to vary with predetermined county characteristics that may be associated with the SNIP program implementation. We add the interactions of birth year dummies and the predetermined county characteristics in 2010, including the number of teachers, proportion of people aged below 15, employment rate, share of the urban population, and GDP per capita. The results are presented in Panel A of Table 5. The estimated effects of exposure to the SNIP have magnitudes close to the baseline estimates and are significant at the conventional level.

Third, we additionally control for province-specific linear trends. Specifically, we add the interactions of province dummies and a linear birth year trend to our baseline specification. The estimates are shown in Panel B of Table 5. The inclusion of the linear trends does not modify our main results.

Fourth, we re-estimate the baseline model with a sample of urban residents. Urban residents shared educational and health care resources with rural residents in the county but were not eligible for the SNIP. This suggests a useful placebo test: the adoption of the SNIP ought to have no impact on children raised in urban families who shared a common socioeconomic environment but were not eligible for the program. Panel C of Table 5 presents the results. The estimated placebo treatment effects are small and statistically not significantly different from zero. It indicates that the estimated effects of the SNIP in rural areas are not driven by other county-specific trends shared by rural and urban residents.⁸ Overall, the four approaches provide supports to our identification strategy.

Other confounding factors – Our estimation strategy evaluates an exogenous policy shock. We, therefore, need to examine whether the estimated effects of the SNIP were confounded by other policies that occurred simultaneously. Several policies may also impact children's cognitive ability and health outcomes. One potential policy candidate is a reform that established a critical illness insurance scheme (CII). The CII aimed to reimburse the high medical expenses associated with serious illness for urban and rural residents (Zhao, 2019). The CII was rolled out across cities between 2012 and 2016. It is possible that the introduction of the CII scheme was correlated with both the introduction of the SNIP and adult outcomes, thereby generating bias in our baseline estimates.

⁶ It is possible to disentangle the effect of the length of the exposure from that of age at first exposure to the school meal program if individuals leave the school program at different ages. With better empirical design and more complete individual-level data, future studies may investigate this issue.

⁷ Some of the dummy variables indicating each specific exposure age are associated with few observations. Hence, we do not have enough powers to detect significant effects for all the different exposure ages.

⁸ We also pool the urban and rural samples and run a triple-difference model. We report the estimates in Table A3 and find qualitatively similar results. It further guards against biases caused by other contemporaneous changes in the county.

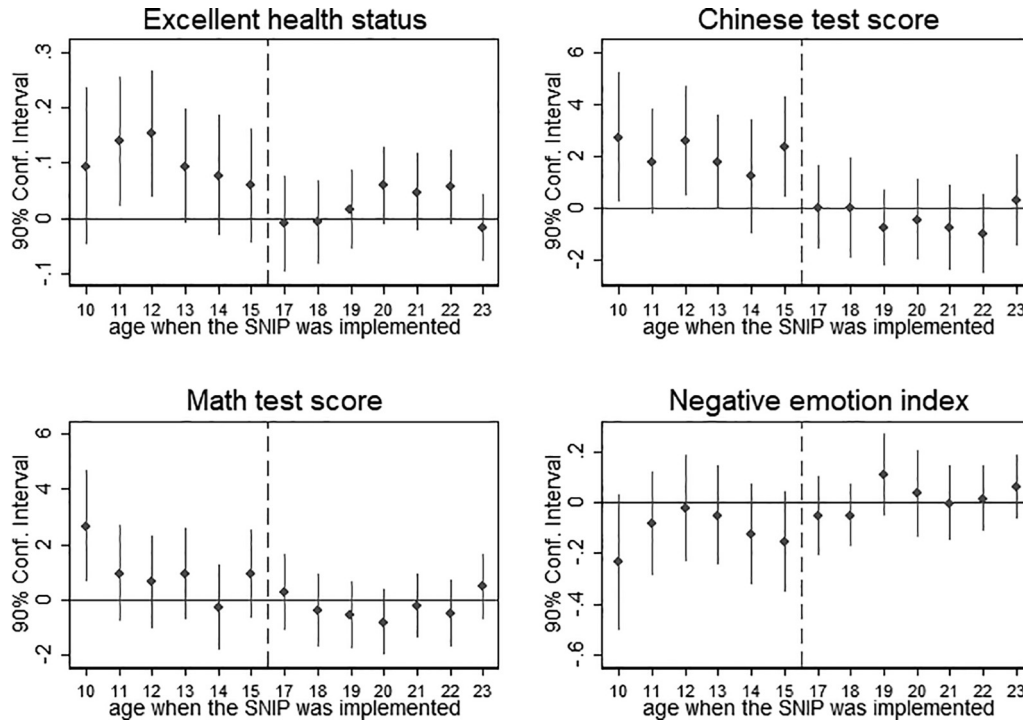


Fig. 1. Event analysis. Note: The figure plots the coefficients from an event-study model. Event time is defined as the age when the SNIP was implemented in the county. The omitted exposure age category is individuals aged above 23 at the implementation of the reform. Vertical lines represent 90% confidence intervals computed from standard errors clustered at the county level. Other control variables are the same as our baseline specification.

Table 5
Confounding factors and sample selection.

	(1)	(2)	(3)	(4)	(5)
VARIABLES	Excellent health status	Negative emotion	Positive emotion	Word test score	Math test score
Panel A Socioeconomic factors					
Reform exposure	0.077** (0.031)	-0.071 (0.059)	-0.029 (0.068)	1.864** (0.736)	0.845* (0.500)
Observations	4,076	3,491	3,491	2,014	2,014
Panel B Province-specific linear trends					
Reform exposure	0.073** (0.030)	-0.107** (0.053)	-0.021 (0.065)	2.851*** (0.910)	1.224** (0.543)
Observations	4,078	3,493	3,493	2,014	2,014
Panel C Urban sample					
Reform exposure	-0.013 (0.081)	0.084 (0.131)	-0.017 (0.141)	0.672 (0.748)	1.139 (0.743)
Observations	746	683	683	445	445
Panel D CII					
Reform exposure	0.072** (0.030)	-0.111** (0.052)	-0.019 (0.065)	2.501*** (0.864)	1.108** (0.493)
Observations	4,078	3,493	3,493	2,014	2,014
Panel E NRPS					
Reform exposure	0.071** (0.030)	-0.112** (0.052)	-0.020 (0.065)	2.510*** (0.874)	1.108** (0.492)
Observations	4,078	3,493	3,493	2,014	2,014
Panel F Migration experience					
Reform exposure	0.076** (0.035)	-0.135** (0.056)	-0.044 (0.072)	2.470*** (0.882)	0.994** (0.499)
Observations	2,843	2,715	2,715	1,845	1,845
Panel G Rural residents					
Reform exposure	0.081** (0.040)	-0.105* (0.060)	0.015 (0.081)	2.918*** (1.041)	1.311** (0.622)
Observations	2,375	1,941	1,941	1,202	1,202

Note: Each cell reports the coefficient of reform exposure from an OLS regression where the dependent variable is the column variable. Reform exposure is defined as a binary variable for being exposed to the SNIP before age 16. Panel A controls for the interactions of birth year dummies and county characteristics in 2010, including the number of teachers, proportion of people aged below 15, employment rate, share of the urban population, and GDP per capita. Panel B controls for the interactions of province dummies and a linear birth year trend. Panel C restricts the sample to urban residents. Panel D controls for a dummy variable for being exposed to the CII before age 16. Panel E controls for a dummy variable for being exposed to the NRPS before age 16. Panel F restricts the sample to those who had not left their hometowns after age 12. Panel G restricts the sample to those who were residing in rural areas at the time of the survey. Control variables include gender, Han ethnicity, parents' community membership, parents' years of education, and fixed effects for county and year of birth. The standard errors are clustered at the county level. ***significant at 1% level, **at 5%, *at 10%.

To test this possibility, we control for a dummy variable that indicates being exposed to the CII scheme before age 16. Data on the timing of the CII scheme comes from local yearbooks, health gazetteers, and government websites. Panel D of Table 5 presents our results with early exposure to the CII controlled for. The estimated coefficients on exposure to the meal program remain similar. Hence, the CII program is an unlikely confounder biasing our estimates.

There is a second possible confounding candidate, which is a pension reform that established the New Rural Pension Scheme (NRPS) in rural China. The NRPS was launched in 320 pilot counties in 2009 and was gradually expanded to cover other counties. The program was voluntary. Rural residents aged 16–59 were eligible to join the program at their places of household registration. Rural residents aged above 60 were entitled to receive benefits without making any contributions at the introduction of the program, as long as all of their adult children participated in the NRPS. It is possible that exposure to the pension reform altered investments in children, thereby affecting their cognitive and health outcomes in the long run (Cheng, Liu, Zhang, & Zhao, 2018; Huang & Zhang, 2021).

To test this idea, we include in our basic regressions an indicator for being exposed to the pension reform before age 14. The estimates are presented in Panel E of Table 5. The estimated coefficients remain mostly unchanged, which suggests minimal bias induced by the pension reform.

Sample selection issue — Concerns have also been expressed that our estimates could be biased by the migration of respondents. First, migration might create measurement errors in the treatment variable. We allocate a county of residence while in school to each respondent by assuming that they were still residing in their county of residence at age 12. Migration during compulsory school age may imply that respondents are assigned to the treatment group even if they should not be and vice versa. This might induce measurement errors and bias our estimates downward. Second, migration might introduce a selection bias. The availability of subsidized school meals might influence choices of residence, especially when not all counties provide subsidized school meals and when few rules regulate school transferring. If the talented or healthy children in the counties which implemented the SNIP were more inclined to migrate before the reform, for example, this could bias our estimates upward.

Fortunately, migration during the compulsory education period was low in China owing to the residence registration system (Yu Chen, Shi, & Tang, 2019). Therefore, a respondent's county of residence at age 12 could be a good approximation of the county of residence during compulsory school age. In the previous analysis, we restrict the sample to respondents who lived in rural areas at age 12 and construct reform exposure based on the implementation year of the SNIP at the respondent's residence at age 12.

To further assess the potential influence of selective migration, we consider two alternative sample restrictions. First, we further limit samples to people who have not left their residing counties since age 12. A total of 1,225 observations are excluded due to the additional restriction. We report the estimates from this sample in Panel F of Table 5. Our conclusions remain mostly unchanged. Second, we restrict the sample to those who live in rural areas at present. A total of 1,703 observations are dropped because of this additional restriction. We report the estimates from this sample in Panel G of Table 5 and find qualitatively similar results. Overall, these tests allay fears that the DID estimates simply pick up the effects of selective migration.

Model specifications — In the baseline analysis, we assume that early exposure to the SNIP is a simple indicator for being exposed to the SNIP before age 16. As a robustness check, we measure early exposure to the SNIP as the number of years exposed to the reform

between ages 6 and 15. Children are supposed to receive compulsory education between ages 6 and 15. Panel A of Table 6 reports the estimates using the linear treatment measure. We continue to find that exposure to the SNIP in childhood improved individuals' health and cognitive outcomes in the long run. The average length of reform exposure conditional on receiving treatment is about 3.2 years in our regression sample. The estimated coefficients indicate that 3.2 years of school meal exposure led to a 7.0 points higher probability of reporting excellent health status, a 0.07-point decrease in the negative emotion index, a 1.76 point increase in word test score, and a 1.06 increase in math test score. The estimated effects are qualitatively similar to those from our binary treatment model.⁹

We then test whether the effect of school meal reform varied with the length of exposure. We use dummy variables for partial and full exposure to quantify the effects of different lengths of reform exposure. Panel B of Table 6 presents the estimates using the alternative measure of reform exposure. The estimates indicate that the school meal reform improved the health and cognitive outcomes of both partially and fully exposed cohorts. Although fully exposed cohorts seem to be affected to a larger extent, the differences are not statistically significant. One possibility is that nutrition in middle schools is vital for growth and development. Another possibility could be that the statistical power was too low to identify significant differences. Overall, we do not have strong evidence for dynamic treatment effects. In addition, the continuous treatment DID model may require a stronger assumption about parallel trends (Callaway, Goodman-Bacon, & Sant'Anna, 2021). Therefore, we use the binary treatment model as our baseline model.

Recent advances in econometric theory highlight the sensitivity of the DID models in staggered treatment settings (e.g. Callaway and Sant'Anna (2021); Goodman-Bacon (2021)). The staggered DID models may give biased estimates of treatment effects when already treated units are used as a control group and the treatment effects are heterogeneous. To examine the extent to which the problem likely matters in our setting, we run the imputation strategy proposed by Borusyak, Jaravel, and Spiess (2021). The key idea of the strategy is to impute potential outcomes for the treated units by using data for the untreated units. A desirable feature of the strategy is its feasibility for both panel and cross-sectional data. The corresponding estimates are presented in Panel C of Table 6. Our main results quantitatively go through under the more robust strategy, which indicates little bias induced by the heterogeneity of the treatment effects.

The CFPS adopted a multi-stage probability sampling strategy to make the sample nationally representative. The 2018 CFPS provided weights for the adult sample. The weighted estimates are reported in Panel D of Table 5. The use of the sampling weights does not qualitatively change the main findings, although it leads to slightly larger standard errors. The use of sampling weight in causal analysis is a controversial issue (Angrist & Pischke, 2008). Weighting may decrease estimation precision when the individual-level error terms are clustered within a unit (Solon, Haider, & Wooldridge, 2015). It is possible that the individual-level error terms within a county are positively correlated with one another in our sample due to unobserved factors at the county level. Therefore, our baseline estimates are unweighted.

A multiple hypothesis testing problem may arise when estimating the effects of the SNIP on multiple outcomes simultaneously. Specifically, the probability of incorrectly rejecting one null hypothesis may be greater than the conventional significance level.

⁹ We have also estimated an augmented model with square and cubed terms of the length of reform exposure. However, most of the estimates are statistically insignificant, which might be caused by the imperfect multicollinearity problem.

Table 6
Model specifications.

	(1)	(2)	(3)	(4)	(5)
VARIABLES	Excellent health status	Negative emotion	Positive emotion	Word test score	Math test score
Panel A Linear reform exposure					
Reform exposure	0.022** (0.009)	−0.021 (0.013)	−0.003 (0.017)	0.550*** (0.200)	0.330** (0.137)
Observations	4,078	3,493	3,493	2,014	2,014
Panel B Step function					
Full reform exposure	0.086* (0.050)	−0.148* (0.078)	−0.052 (0.086)	2.424** (0.997)	1.736** (0.739)
Partial reform exposure	0.069** (0.030)	−0.103* (0.055)	−0.012 (0.068)	2.517*** (0.913)	0.926* (0.504)
Observations	4,078	3,493	3,493	2,014	2,014
F statistics	0.161	0.339	0.247	0.0122	1.500
P-values	0.689	0.562	0.620	0.912	0.223
Panel C Staggered DID					
Reform exposure	0.075** (0.035)	−0.120** (0.050)	−0.035 (0.066)	2.376*** (0.714)	1.120** (0.451)
Observations	4,078	3,493	3,493	2,014	2,014
Panel D Weighted estimates					
Reform exposure	0.078* (0.046)	−0.085 (0.063)	−0.064 (0.069)	1.821** (0.872)	1.239** (0.551)
Observations	4,078	3,493	3,493	2,014	2,014

Note: Each cell reports the coefficient of reform exposure from an OLS regression where the dependent variable is the column variable. Panel A measures reform exposure by the number of years that the respondent was exposed to the SNIP at ages 6–15. Panel B measures reform exposure by two dummy variables. Full exposure is an indicator for being exposed to the SNIP before age 12. Partial exposure is an indicator for being at first exposed to the SNIP between ages 12–15. The F statistics and the corresponding p-values reported in Panel B test whether the coefficients on partial exposure and full exposure are equal. Panel C uses the imputation strategy proposed by [Borusyak et al. \(2021\)](#). Panel D reports the weighted estimates. Control variables include gender, Han ethnicity, parents' community membership, parents' years of education, and fixed effects for county and year of birth. The standard errors are clustered at the county level. ***significant at 1% level, **at 5%, *at 10%.

We use the adaptive linear method proposed by [Benjamini, Krieger, and Yekutieli \(2006\)](#) to adjust for simultaneous inferences. The method controls the false discovery rate (FDR), which measures the likelihood of type I errors in null hypothesis testing. The FDR-adjusted p-values are reported in Panel A of [Table A4](#). The estimates of interest are still significant at the conventional level, which demonstrates minimal bias caused by the multiple hypothesis testing problem.

5.3. Mechanism analysis

As mentioned previously, there are several plausible explanations for why early-life nutrition intervention improved later-life cognitive and health outcomes. We classify them into two broad categories: contemporaneous improvement in nutrient intakes and household economic conditions. In this section, we use the 2010–2018 waves of the CFPS to discuss these potential explanations.

We begin by analyzing the short-term effects of the SNIP reform on children's nutrient intakes. Columns (1) to (4) of [Table 7](#) report the corresponding estimates. The core independent variable is a dummy variable indicating whether the reform was implemented in the county of residence. The dependent variables are dummy variables indicating whether the respondent consumed certain types of foods in the last week. As expected, the SNIP improved children's nutrient intakes. Specifically, the SNIP significantly increased the probability of consuming eggs, milk, and beans by at least 6.2 percentage points. These foods provided high-quality protein, zinc, calcium, and other micronutrients, which could improve children's health status and promote cognitive development. Although the SNIP seemed to increase the consumption of fish and meat, the estimated coefficient is small and statistically insignificant.

One might wonder whether the improved nutrient intakes led to overweight. Columns (5) and (6) of [Table 7](#) present the estimates for being overweight and underweight. We find no significant effects of the SNIP on the probability of being overweight and underweight. The school meal program did not influence body size,

suggesting that the program did not provide an unhealthy excess of calories.

The school meal program might also impact family economic outcomes. In traditional Chinese families, mothers are primarily responsible for family care and household chores. According to the 2010 CFPS survey, about 50% of mothers with children aged 6–15 spent more than four hours per day on family care and housework, and 20% spent more than six hours per day on family care and housework. The burden of domestic responsibilities significantly decreased the labor force participation rate and earnings of women and the effects were stronger for rural women ([Ye & Zhao, 2018](#)). Without the subsidized school meal program, mothers might have to provide lunch at home.¹⁰ The introduction of the school meal program might ease the burden of household work for mothers and improve their labor market performance. Columns (1) and (2) of [Table 8](#) reports the estimates for mothers' employment and income, respectively. As expected, the introduction of the SNIP significantly increased the probability of mothers' employment by 4 percentage points and raised mothers' income by 17.5%.

Owing to the costs of caring for young children and traditional gender roles, mothers may not be able to realize their labor market potential. Numerous countries have implemented childcare subsidy policies to relax the burden of childcare. A growing number of recent studies have documented that childcare subsidy policies improved mothers' labor market performance (see [Cascio, Haider, and Nielsen \(2015\)](#), [Akgunduz and Plantenga \(2018\)](#) for recent reviews). The studies on developing countries are still limited. [Berlinski and Galiani \(2007\)](#) assessed the effect of an extensive project constructing pre-primary school facilities in Argentina and found that one preschool facility constructed per ten children at preschool age increased maternal employment by 1.4 percentage points. [Du and Dong \(2013\)](#) found that the presence of day care centers in the community increased the maternal labor force participation rate by 10.5 percentage points in China. [Martínez and](#)

¹⁰ According to the 2011 wave of the China Health and Nutrition Survey, about 85% of mothers aged 25–55 prepared food last week but only 28% of fathers aged 25–55 prepared food last week.

Table 7
Nutritional status of children.

	(1)	(2)	(3)	(4)	(5)	(6)
VARIABLES	Fish and meat	Beans	Milk and eggs	Vegetable and fruit	Underweight	Overweight
Treatment	0.019 (0.049)	0.087** (0.039)	0.062* (0.036)	0.031 (0.028)	−0.015 (0.021)	0.009 (0.014)
Observations	5,116	5,116	5,116	5,116	13,464	13,464
R-squared	0.323	0.179	0.184	0.115	0.062	0.059
County FE	YES	YES	YES	YES	YES	YES
Survey year FE	YES	YES	YES	YES	YES	YES

Note: The core independent variable is a dummy variable indicating whether the SNIP was implemented in the county of residence. Other control variables include gender, Han ethnicity, parents' community membership, and parents' years of education. The standard errors are clustered at the county level. ***significant at 1% level, **at 5%, *at 10%.

Table 8
Family economic outcomes.

	(1)	(2)	(3)	(4)	(5)	(6)
VARIABLES	Mother's employment	Mother's income	Father's employment	Father's income	Educational expenditure	Food expenditure
Number of treated children	0.040** (0.019)	0.175** (0.072)	0.027 (0.019)	0.060 (0.041)	0.233*** (0.085)	−0.038 (0.030)
Observations	8,518	3,027	8,126	4,288	10,951	10,801
R-squared	0.183	0.360	0.167	0.329	0.141	0.158
County FE	YES	YES	YES	YES	YES	YES
Survey year FE	YES	YES	YES	YES	YES	YES

Note: The core independent variable is the number of children affected by the SNIP program. Other household-level control variables include the average age of children, sex ratio of children, and ages of parents. The standard errors are clustered at the county level. ***significant at 1% level, **at 5%, *at 10%.

Perticar  (2017) conducted a field experiment in Chile and observed that the provision of afterschool care for children aged between 6 and 13 raised maternal employment by 5 percentage points but had no obvious effect on income. The estimated effects of the school meal reform in our setting are comparable to those from the previous studies, which suggests that the preparation of meals for children might be a major impediment to maternal employment.

Fathers are usually assumed to be breadwinners of households. They often undertake a small portion of household chores, including the preparation of children's meals (Ye & Zhao, 2018). According to the 2010 CFPS survey, 12% of fathers with children aged 6–15 spent more than four hours and 4% spent more than six hours per day on family care. Therefore, we expect the school meal program had smaller effects on the labor supply of fathers. Columns (3) and (4) of Table 8 presents the estimates for fathers' employment and income. Consistent with our expectations, the school meal program had small and statistically insignificant effects on fathers' employment and income.

Furthermore, we examine the short-term effects of the SNIP on household food expenditure and educational investments. Columns (5) and (6) of Table 8 presents the corresponding estimates. The estimates indicate that the SNIP significantly increased household educational investments but had an insignificant negative effect on household food expenditure. Although subsidized school meals reduced children's food expenditure at schools, it might increase other types of household food expenditure due to increased household income. Therefore, the overall effect of household food expenditure was small and statistically insignificant.¹¹

5.4. Heterogeneity analysis

We have shown that individuals exposed to the SNIP program during the compulsory school period experienced significant

improvements in cognitive and health outcomes. An important question is whether these improvements were heterogeneous across subpopulations. We examine this issue in this subsection.

The long-run effects of the school meal program may differ by gender. Panels A and B of Table 9 report the effects of the SNIP separately for males and females. Early exposure to the SNIP had significantly positive effects on health outcomes for males whereas the estimated effects for females are small and insignificant. This discrepancy may be related to gender differences in energy requirements. In general, males require more energy to maintain their body function and daily life than females do (Sorhaindo & Feinstein, 2006). Consequently, males may obtain larger health benefits from the school nutrition program. With respect to cognitive outcomes, early exposure to the SNIP seemed to have larger and more significant effects on math and word test scores for females. A potential explanation is that girls received much fewer educational investments than boys did in rural China (Golley & Kong 2018; Song & Zhou 2019). Preference for sons is common in rural China and parents usually give preferential treatment toward boys. According to the 2010 census data, the proportions of males and females aged 20–29 who had not finished compulsory education were 7.4% and 10.5%, respectively. The school meal program generated more obvious effects on cognitive test scores for females because their learning potentials were not fully realized.

The effects of the school meal program may depend on family socioeconomic status. Parents with low socioeconomic status usually faced binding financial constraints and had inadequate nutrition knowledge. Their children have a high likelihood of consuming low-quality diets and suffering from nutritional deficiencies, and thereby are susceptible to the metabolic changes that influence cognitive performance and health status (Afridi, 2010). Nutritious school meals might augment their nutrient intake and lead to improved performance. By contrast, children from high socioeconomic status families are usually well-nourished. Therefore, we suspect that the benefits of subsidized school meals are more pronounced for respondents from low socioeconomic status families.

We then evaluate the heterogeneous effects of the school meal program across socioeconomic status. A family's socioeconomic

¹¹ Our mechanism analysis also includes multiple outcomes. We also use the adaptive linear step-up method to adjust for simultaneous inferences. We report the FDR-adjusted p-values in Panels B and C and find similar results.

Table 9
Heterogeneity analysis.

	(1)	(2)	(3)	(4)	(5)
VARIABLES	Excellenthealth status	Negativeemotion	Positiveemotion	Wordtest score	Mathtest score
Panel A Male					
Reform exposure	0.113*** (0.040)	−0.158** (0.077)	0.021 (0.085)	1.746* (0.944)	0.351 (0.617)
Observations	2,133	1,817	1,818	989	989
Panel B Female					
Reform exposure	0.025 (0.038)	−0.062 (0.070)	−0.061 (0.081)	3.339*** (1.172)	1.634** (0.705)
Observations	1,936	1,667	1,666	1,010	1,010
Panel C Less-educated fathers					
Reform exposure	0.141*** (0.046)	−0.162* (0.087)	0.057 (0.109)	4.043*** (1.296)	1.340 (0.952)
Observations	1,765	1,455	1,455	839	839
Panel D Highly-educated fathers					
Reform exposure	0.025 (0.040)	−0.067 (0.065)	−0.068 (0.073)	0.898 (1.108)	0.603 (0.592)
Observations	2,299	2,023	2,023	1,156	1,156

Note: Each cell reports the coefficient of reform exposure from an OLS regression. All dependent and independent variables are the same as our baseline specification. Panels A to D report the estimates for male, female, individuals with less-educated fathers, and individuals with highly-educated fathers, respectively. The standard errors are clustered at the county level. ***significant at 1% level, **at 5%, *at 10%.

status is usually proxied by the father's educational level. The respondents were divided into two groups based on whether or not their fathers received at least middle school education. Panels C and D of Table 9 show the estimates for individuals with highly- and less-educated fathers, respectively. As expected, early exposure to the SNIP mostly had positive effects on cognitive and health outcomes of individuals with less-educated fathers whereas the coefficients for respondents with highly-educated fathers were usually small and insignificant.

6. Conclusion

School meal programs are popular interventions to fight child malnutrition in developing countries (Jomaa et al., 2011). This paper examines the long-term impacts of early exposure to the school meal program on adult cognitive and health outcomes in China. The county-by-county rollout of the SNIP program across counties created exogenous variation in cumulative exposure to the SNIP. Analyses of the CFPS suggest that exposure to the SNIP before age 16 was associated with statistically significant and meaningful improvements in cognitive ability and self-reported health status. These effects are only observed in rural residents targeted by the program. We provide some suggestive evidence that the school meal program immediately increased children's nutrient intakes, improved mothers' labor market performance, and raised households' investments in education. We find no evidence for a contemporaneous increase in the likelihood of being overweight. Children from poor households gained larger benefits from the program than children from wealthy households, suggesting that the program decreased socioeconomic inequalities in adulthood.

Our results suggest that the school nutrition program could be an important investment in children's long-run human capital. These results may be relevant for other developing countries. The SNIP program was introduced in rural areas where the infrastructure supply was poor and the living standard of people was low. The program changed the nutrient intakes of school-age children and made children switch away from low-quality lunch bags or lunches at home. The program also released mothers from the task

of providing school meals and improved their labor market performance. This may be relevant for many developing countries today that attempt to provide or have provided subsidized school meals.

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Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Appendix

Tables A1–A4.

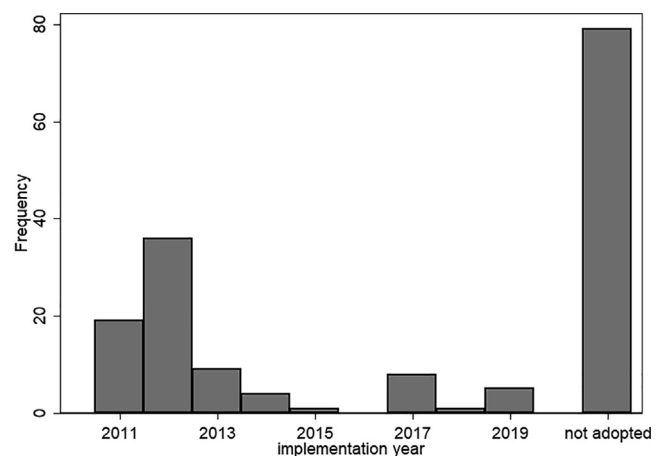


Fig. A1. Implementation years of the SNIP program. Note: The figure plots the frequency of implementation years of the SNIP program for the 162 counties surveyed by the CFPS.

Table A1
Other outcomes.

	(1)	(2)	(3)	(4)
VARIABLES	Health statusindex	Standardized wordtest score	Standardized mathtest score	CES-D scale
expos1	0.152** (0.066)	0.233** (0.103)	0.171* (0.092)	−0.172 (0.250)
Observations	4,078	2,014	2,014	3,490
R-squared	0.081	0.291	0.306	0.084
Birth year FE	YES	YES	YES	YES
County FE	YES	YES	YES	YES

Note: The health status index is an ordered variable where higher values represent better health status. Word and math cognitive test scores are standardized by age. The CES-D scale is a measure of depressive symptoms, with higher values indicating higher risks for clinical depression. The core independent variable is a binary variable indicating whether the respondent was exposed to the SNIP program before age 16. Other control variables include gender, Han ethnicity, parents' community membership, and parents' years of education. The standard errors are clustered at the county level. ***significant at 1% level, **at 5%, *at 10%.

Table A2
Mental health outcomes.

	(1)	(2)	(3)	(4)	(5)
VARIABLES	Lonely	Sad	Sleepless	Enjoying life	Happy
Reform exposure	−0.150** (0.064)	−0.126** (0.053)	0.074 (0.073)	0.035 (0.074)	0.048 (0.085)
Observations	1,942	1,941	1,942	1,948	1,949
R-squared	0.082	0.083	0.091	0.101	0.093
Birth year FE	YES	YES	YES	YES	YES
County FE	YES	YES	YES	YES	YES

Note: The dependent variables are five ordered variables measuring mental health status of respondents in the last month. Specifically, the respondents evaluated on a four-point scale (1) the frequency of feeling lonely, (2) the frequency of feeling sad, (3) the frequency of poor sleep, (4) the frequency of enjoying life, and (5) the frequency of feeling happy. The core independent variable is a binary variable indicating whether the respondent was exposed to the SNIP program before age 16. Other control variables include gender, Han ethnicity, parents' community membership, and parents' years of education. The standard errors are clustered at the county level. ***significant at 1% level, **at 5%, *at 10%.

Table A3
Triple-difference estimation.

	(1)	(2)	(3)	(4)	(5)
VARIABLES	ExcellentHealth status	Negativeemotion	Positiveemotion	WordTest score	MathTest score
Reform exposure*Rural area	0.079** (0.040)	−0.102* (0.060)	0.014 (0.081)	2.951*** (1.042)	1.326** (0.618)
Observations	4,832	4,183	4,183	2,474	2,474
R-squared	0.096	0.100	0.119	0.392	0.437
Birth year FE	YES	YES	YES	YES	YES
County FE	YES	YES	YES	YES	YES

Note: The key explanatory variable is the interaction term of the reform exposure measure and the dummy for rural areas. Other control variables include indicators for rural areas and exposure age, as well as other baseline covariates.

Table A4
Adjusted statistical inference.

	(1)	(2)	(3)	(4)	(5)
VARIABLES	Coefficients of interest	Standard errors	Unadjusted p-values	FDR-adjusted p-values	Observations
Panel A Baseline outcomes					
Excellent Health status	0.072**	(0.030)	0.018	0.035	4,078
Negative emotion	−0.112**	(0.052)	0.033	0.035	3,493
Positive emotion	−0.019	(0.065)	0.768	0.182	3,493
Word Test score	2.497***	(0.865)	0.005	0.024	2,014
Math Test score	1.105**	(0.494)	0.027	0.035	2,014
Panel B Nutrition status					
Fish and meat	0.019	(0.049)	0.691	0.856	5,116
Beans	0.087**	(0.039)	0.026	0.099	5,116
Milk and eggs	0.062*	(0.036)	0.090	0.292	5,116
Vegetable and fruit	0.031	(0.028)	0.271	0.568	5,116
Underweight	−0.015	(0.021)	0.483	0.707	13,464
Overweight	0.009	(0.014)	0.518	0.707	13,464
Panel C Economic outcomes					
Mother's employment	0.040**	(0.019)	0.044	0.063	8,518
Mother's income	0.175**	(0.072)	0.017	0.045	3,027
Father's employment	0.027	(0.019)	0.152	0.101	8,126
Father's income	0.060	(0.041)	0.137	0.101	4,288
Educational expenditure	0.233***	(0.085)	0.007	0.045	10,951
Food expenditure	−0.038	(0.030)	0.208	0.116	10,801

Note: Each line reports the estimation result from an OLS regression where the dependent variable is the row variable. The specifications are the same as our baseline and mechanism analysis. The core independent variables in Panel A, B, and C are reform exposure, treatment, and the number of treated children, respectively. The adaptive linear method proposed by Benjamini et al. (2006) is conducted separately for each panel. The standard errors are clustered at the county level. Columns (3) and (4) report the conventional and FDR-adjusted p-values, respectively. *** significant at 1% level, ** at 5%, * at 10%.

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